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Accumulation and translocation of methylmercury and inorganic mercury in *Oryza sativa*: An enriched isotope tracer study

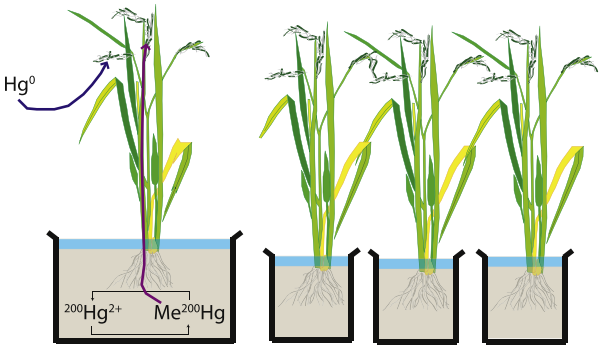
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HIGHLIGHTS

- Rice (*Oryza sativa*) was experimentally grown in sediment dosed with enriched ^{200}Hg.
- Mercury methylated in sediment was the sole source of methylmercury in rice.
- *In planta* demethylation in rice plants is likely, but *in planta* methylation is not.
- Inorganic Hg in rice grain was almost entirely from an atmospheric source.

GRAPHICAL ABSTRACT



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ABSTRACT

Methylmercury (MeHg) accumulation in rice is an emerging human health issue, but uptake pathways and translocation into the grain remain poorly understood. We grew *Oryza sativa* plants in pots of wetland soil amended with an enriched mercury isotope (94.3% ^{200}Hg) tracer, alongside unvegetated control pots, and assessed both ambient and tracer MeHg and inorganic Hg (IHg) concentrations in soil and plant tissues at three growth stages. Based on similar ratios of ambient:tracer MeHg concentrations in soil and plant tissues, we provide the first direct evidence that MeHg is first synthesized in saturated soil and subsequently translocated to rice grains. There is no evidence of *in planta* methylation of IHg, but significant losses of MeHg from plant tissues between flowering and maturity indicates likely *in planta* demethylation. In this greenhouse experiment, lower percent of tracer MeHg in vegetated soils at late growth stages suggests that rice plants reduce the net MeHg accumulation capacity of soils, although the mechanism remains unclear. For IHg, roots accumulated Hg from the soil, straw from the soil and the atmosphere, and grain almost entirely from the atmosphere. Management strategies that aim to reduce MeHg accumulation in rice should focus on mercury methylation in paddy soils, but IHg reductions will depend on regional controls of atmospheric Hg.

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1. Introduction

Methylmercury (MeHg) is the highly bioaccumulative form of mercury (Hg) and is a contaminant of global concern due to links between low level exposure and neurological impairment (Clarkson

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and Magos, 2006). Exposure to MeHg is mostly associated with the consumption of aquatic organisms, particularly ocean fish (Li et al., 2014), but ingestion of rice cultivated in Hg-contaminated areas of China (Feng et al., 2008; Li et al., 2012; Rothenberg et al., 2011b; Zhang et al., 2010a), Cambodia (Cheng et al., 2013), and elsewhere (Rothenberg et al., 2014) has been gaining attention as an additional, significant source of dietary MeHg. Since rice is the world's most widely cultivated crop, and the staple food of approximately 3 billion people (Khush, 2005), MeHg exposure via rice consumption is a growing concern worldwide.

Like other wetlands, flooded rice paddies are sites of active Hg methylation (Liu et al., 2014; Rothenberg and Feng, 2012), a process carried out by a wide taxonomic range of anaerobic microorganisms (Gilmour et al., 2013). Correlations between soil and rice grain MeHg concentrations suggest that grain MeHg originates from Hg methylated in the soil (Meng et al., 2010; Zhang et al., 2010b); however, it is also possible that inorganic Hg (IHg) may be methylated *in planta* (Horvat et al., 2003; Rothenberg et al., 2014). The source(s) of IHg to the rice plant are also uncertain. There is a strong association between IHg and MeHg in rice grains (Rothenberg et al., 2014), but several authors have suggested that translocation pathways and sources may differ between IHg and MeHg. IHg in rice plants may be absorbed from soil (Zhang et al., 2010b), the atmosphere (Meng et al., 2011), or a mixture of these sources (Meng et al., 2014, 2012), with the dominant source likely differing among plant tissues (Meng et al., 2010; Yin et al., 2013). Without a robust understanding and greater certainty of where MeHg and IHg in rice grain originates, effective engineering or environmental management strategies to mitigate MeHg contamination of rice grain cannot be successfully implemented.

The current understanding of both MeHg and IHg accumulation in rice is mostly based on correlative research in which Hg or MeHg concentrations in different plant and environmental compartments are related to one another, and translocation pathways are inferred. Enriched stable isotope tracers are a powerful alternative to this correlative approach (Werner et al., 2012), and have been applied to understanding the sources, translocation, and *in planta* distribution of Hg through forest understory vegetation (Schwesig and Krebs, 2003) and sawgrass (Mao et al., 2013), but not rice. In this study, we combined enriched stable isotope tracer techniques with a controlled greenhouse experiment cultivating rice plants to investigate the following questions: 1) Is MeHg translocated from soil to aboveground plant parts (including grain) and, if so, when does this occur? and 2) Does IHg originate from similar sources as MeHg?

2. Experimental

2.1. Experimental design

This greenhouse rice cultivation experiment was conducted as a complete, randomized block design incorporating two vegetation treatments (rice/no rice) and three growth stages, with ten replicates for each combination. All pots of soil were dosed with an enriched stable Hg isotope tracer (94.3% enriched ^{200}Hg). Rice (*Oryza sativa* L. ssp. *japonica* cv. Nipponbare) was cultivated in flooded pots under greenhouse conditions and unvegetated pots were maintained as controls. At tillering, flowering, and grain maturity growth stages, ten plants/pots were sacrificed and soil, stems and leaves (collectively straw), roots, panicles, and grain were sampled, as applicable. Tillering, characterized by the formation of lateral shoots, signalled the initiation of rapid vegetative growth; flowering represented the beginning of reproduction; and grain maturity was assessed by the presence of drooping, filled grains. At the same time as vegetative sampling, soil was sampled from unvegetated control pots. All samples were simultaneously analysed for natural background (ambient) and excess ^{200}Hg (tracer) concentrations of total Hg (THg) and MeHg.

2.2. Sediment dosing, plant cultivation and sample collection

Sediment was collected from the top fifty centimeters of a shallow, vegetated littoral area of Tangamong Lake (44°42′46″ N 77°50′12″ W), near Havelock, Ontario, Canada, in May of 2015. Soil was homogenized by hand and fragments of plant material and detritus were removed, resulting in an organic matter content of 12.8% and a bulk density of 0.388 g cm^{-3}. Sixty liters of soil were amended with an enriched (94.3%) ^{200}Hg solution (as $HgCl^{+}$) to arrive at a final isotope tracer concentration of approximately 50 ng g^{-1} dry weight and kept in the dark at 4 °C for one month to allow the added Hg tracer to adsorb to and equilibrate with the soil. During this time, soils were fully saturated and stirred daily to prevent the formation of strongly reducing conditions. After one month, 1.5 L of soil was added to each of forty opaque, watertight three-liter plastic pots. The pot filling order was randomized, as was the assignment of pots to blocks ($n = 10$). Blocks were re-randomized weekly.

Rice seeds (*Oryza sativa* L. ssp. *japonic*a cv. Nipponbare) were obtained from the Genetic Stocks-Oryza collection of the United States Department of Agriculture in Stuttgart, Arkansas. Seedlings were germinated and pre-cultivated hydroponically to reduce physiological variability between replicates. Seeds were rinsed thoroughly in deionized water and germinated in aerated distilled water, and two seeds each sown in 0.5-L plastic containers filled with a 50–50% soilless mixture of perlite and vermiculite. Seedlings were grown for 25 days under ambient laboratory light supplemented with sodium lamps on a 12 h–12 h light-dark cycle in an aerated hydroponic solution with the following nutrient composition: N: 165 mg L^{-1}, P: 50 mg L^{-1}, K: 223 mg L^{-1}, Ca: 313 mg L^{-1}, Mg: 120 mg L^{-1}, S: 122 mg L^{-1}, Fe: 7 mg L^{-1}, Mn: 0.2 mg L^{-1}, Zn: 0.4 mg L^{-1}, Cu: 0.1 mg L^{-1}, B: 1.3 mg L^{-1}, Mo: 0.06 mg L^{-1}. The nutrient solution was aerated with two bubblers repositioned daily, adjusted to pH 7.5 with 0.5 M HCl and NaOH as necessary, and refreshed weekly. On day 15, duplicate seedlings were removed, leaving a single plant per container. When the seedlings had reached approximately 15 cm in height on Day 23, seedling height was measured and their photosynthetic rate was assessed as a proxy measure of physiological status using a portable photosynthesis system (Licor-6400XT) with the following internal parameters: 385 ppm carbon dioxide concentration, 1200 μmol m^{-2} s^{-1} photosynthetic photon flux density, and 40% relative humidity. Finally, thirty seedlings were selected on the basis of similar height and photosynthetic rate, randomly assigned to treatments, and transplanted to soil filled pots. To maintain flooded conditions, water was added as necessary to a level of 5–6 cm above the soil surface. Plants were grown in the greenhouse facilities of the University of Toronto at Scarborough. Natural light was supplemented with sodium lamps on a 12 h–12 h day-night cycle with a three-week dark treatment (10 h–14 h day-night) beginning 77 days after germination to hasten flowering. Temperature was maintained at 25–30 °C. All pots were fertilized every 10 days with alternating applications of 239.65 g L^{-1} Peters Unimix water soluble fertilizer (NPK 15–5-15; Solution A), and 239.65 g L^{-1} Peters Unimix Plus with 48 g L^{-1} Dissolvine chelated iron supplement (Solution B) until flowering, after which only Solution A was applied (schedule modified from Xu et al., 2009). Unvegetated pots received the same schedule of fertilizers.

Sampling campaigns and plant harvests were conducted when all plants had reached the appropriate growth stage, which occurred 67 days after germination for tillering, 91 days after germination for flowering, and 113 days after germination for grain maturity. At each growth stage, ten vegetated pots were sacrificed. A small plastic soil corer was used to collect a sample of the entire soil profile, extending from the top of the soil to the bottom of the pot, from both vegetated pots and the unvegetated control pots. Photosynthetic rates were measured in the greenhouse using the same Licor instrument and settings described above. The rice plants were then removed from the pots and transported back to the lab in individually labelled plastic bags, where the entire plant was separated into aboveground and belowground

biomass. Soil was carefully removed from belowground biomass (roots) by copious washing with tap water, removal of large debris, and a final rinsing with deionized water. Aboveground biomass was divided into reproductive and nonreproductive fractions as applicable (stems/leaves, collectively straw; panicles; and grain). Grain husks were removed by hand. All plant fractions were weighed after harvest and the dry mass calculated from subsamples that were dried at 104 °C until stable mass was achieved. Soil and plant tissues for Hg and MeHg analysis were freeze-dried and crushed in a glass mortar or shredded with stainless-steel scissors to a fine powder, and stored in airtight plastic containers until analysis. Homogenization tools were thoroughly cleaned between samples to prevent cross-contamination.

2.3. Hg analyses

Samples were simultaneously analysed for both ambient MeHg (using the dominant ^{202}Hg isotope) and excess Me^{200}Hg (tracer; attributed to the added tracer after accounting for ambient isotope ratios) according to Hintelmann and Evans (1997). For MeHg analysis, we applied species-specific isotope dilution techniques wherein samples were distilled in a solution of deionized water, H_2SO_4, KCl, and $CuSO_4$ in acid-cleaned Teflon vessels to which a known mass of enriched Me^{199}Hg was added as an internal standard, with analysis by gas chromatography inductively coupled plasma mass spectrometry (ID-GC-ICPMS; Hintelmann et al., 1997). Briefly, MeHg was liberated from solution via aqueous phase ethylation using sodium tetraethylborate, purged by bubbling with high-purity nitrogen, and concentrated onto a glass trap filled with Tenax®. Tenax traps were thermally desorbed on a stream of argon gas, directed through a chromatography column for separation of species and ultimately into an ICP-MS (Agilent Technologies 7700x) for Hg isotope detection. For THg analysis, 0.2–0.3 g of sample was digested in 10 mL of hot nitric acid until vapours became colourless. Digestates were diluted with deionized water to 50 mL volume, oxidized to Hg^{2+} by the addition of 0.5 mL BrCl, and left to react overnight. Digestates were further diluted with deionized water and run on a Tekran 2600 automated Hg analysis system (automated $SnCl_2$ reduction, dual gold trap amalgamation, and thermal desorption) that was directly hyphenated to an ICP-MS. Inorganic Hg concentrations in each sample were obtained by subtracting MeHg concentrations from THg concentrations. Ongoing quality control was assessed by the analysis of blanks, duplicate samples, certified sediment reference materials (MESS-3 for THg; ERM CC580 for MeHg) and matrix spikes (THg only). Method detection limits were based on three times the standard deviation of blank results, corrected for the average mass of sample digested. These data are displayed in Table 1.

2.4. Statistical analyses

For each variable, normality was assessed separately for soil, roots, straw/panicles, and grain using quantile-quantile plots and frequency histograms. Non-normal variables were successfully transformed. Variables were compared using ANOVA or one- and two-way T-tests at $\alpha \leq 0.05$. All tests were performed using RStudio V 0.99.484. The results of all statistical tests are available in Supplementary Table 1.

Table 1
Quality control data for MeHg and THg determinations.

	Method detection limit (ng g^{-1})	Certified reference material recovery (%)	Average duplicate RSD (%)	Spike recovery (%)
Ambient MeHg	0.05	100 ± 11 (n = 32)	5.8 ± 5.2 (n = 26 pairs)	
Excess Me^{200}Hg	0.08		6.3 ± 13.4 (n = 18 pairs)	
Ambient THg	0.42	101 ± 3 (n = 12)	4.2 ± 3.0 (n = 14 pairs)	95 ± 11 (n = 6)
Excess T^{200}Hg	0.11		7.0 ± 5.6 (n = 10 pairs)	

3. Results and discussion

3.1. Inorganic mercury tracer amendment

Tracer additions closely matched ambient soil IHg concentrations. In preplanting soils, concentrations of both ambient (mean ± standard deviation = 59.95 ± 18.91 ng g^{-1}) and tracer (48.60 ± 16.27 ng g^{-1}) IHg were at the low end of the background soil Hg levels in the U.S. (Obrist et al., 2011) and even in combination, well below the level of IHg observed to affect soil ecology (Vishnivetskaya et al., 2010). The ratio of ambient:tracer IHg concentrations in vegetated and unvegetated treatments did not vary significantly over the three sampling points (1.19 ± 0.15–1.23 ± 0.13; Supplementary Table 2) from the ambient:tracer IHg ratio in preplanting amended soil (ambient:tracer ratio of 1.24 ± 0.05) (Fig. 1a and Supplementary Table 1; $p > 0.05$). Thus, sediment homogenization had controlled potential variation in tracer additions between experimental replicates. Tracer Hg reduced in the soils and evaded to the greenhouse atmosphere could not possibly have been a detectable secondary source of Hg to the plants. Based on the air volume and turnover of the greenhouse, area of the experimental pots, published rates of Hg evasion from vegetated wetlands (Lindberg et al., 2005; Marsik et al., 2005), and the background range of Hg in the air of the Northern Hemisphere (Driscoll et al., 2013), we estimate that possible evasion of soil tracer Hg to the greenhouse atmosphere would have represented considerably <1% of the ambient Hg present in the air (see Supplementary Text 1 for calculations). We are therefore confident that any tracer Hg detected as present in plant tissues can only possibly be from a source in the soil.

3.2. Mercury methylation in vegetated and unvegetated soils

Ambient MeHg concentrations in the soil were relatively low throughout the experiment in both unvegetated (overall mean across all growth stages: 0.59 ± 0.07 ng g^{-1}) and vegetated (0.52 ± 0.09 ng g^{-1}) soils. Tracer MeHg concentrations were consistently and significantly higher than ambient MeHg concentrations in unvegetated (1.61 ± 0.26 ng g^{-1}) and vegetated (1.21 ± 0.49 ng g^{-1}) soils at all time points (Supplementary Tables 1, 2). Based on the ambient and tracer IHg levels present in preplanting soils, a ratio of 1.24 ambient:tracer MeHg concentrations would be expected if the tracer were methylated at a similar rate to the Hg already present in the soil. In contrast, ambient:tracer MeHg ratios in sediments from tillering, flowering, and maturity growth stages ranged from 0.36 ± 0.04–0.44 ± 0.09 (Fig. 1b), representing a ~2.6 times greater bioavailability of the tracer for methylation, a finding in concurrence with previous Hg amendment experiments (Harris et al., 2007; Mao et al., 2013; Peng et al., 2012; Schwesig and Krebs, 2003). One possible explanation for this increased mercury methylation may be the greater bioavailability of the newly deposited Hg, which some authors have suggested is mostly bound in nanoparticulate (<0.45 μm) Hg sulphide (HgS) particles (Zhang et al., 2012). As Hg ages, nanoparticles may coalesce into large colloids or crystalline HgS compounds, rendering it less available for uptake by Hg methylating microorganisms (Graham et al., 2012). While further research is required into why newly deposited Hg is more microbially bioavailable and for how long this effect endures, this difference should not impact MeHg uptake mechanisms or our interpretation of sources.

The major intent of our study was to assess pathways of MeHg uptake in rice; however, the dosing of soil with enriched Hg isotope and inclusion of unvegetated control pots also allowed for a unique controlled experiment to assess whether wetland plants such as rice have a positive or negative effect on mercury methylation in rhizosphere soil. Field-based studies have repeatedly observed positive relationships

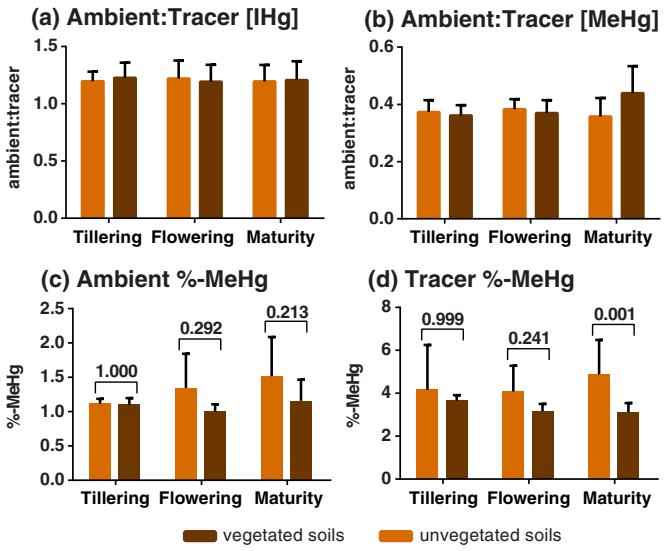


Fig. 1. Mercury in vegetated and unvegetated soils. Top: ratios of ambient to tracer concentrations for IHg (a) and MeHg (b). Bottom: Percent of mercury present as MeHg (%-MeHg) for ambient (c) and tracer (d). Results of ANOVA post-hoc tests appear above the relevant pair of columns. All bars display mean ± standard deviation.

between MeHg accumulation and the presence and activity of plants in both natural (Canário et al., 2007) and agricultural (Windham-Myers et al., 2009) wetlands, strongly suggesting that MeHg accumulation in vegetated pots should have been higher than in unvegetated pots. However, our controlled greenhouse study did not support the hypothesis that plants and/or their rhizosphere enhance Hg methylation. When we examined the percentage of Hg present as MeHg (%-MeHg), a proxy of net MeHg production that integrates demethylation and reflects the long-term net MeHg accumulation capacity of a soil (Drott et al., 2008), we observed that %-MeHg was consistently lower in vegetated than unvegetated pots (Fig. 1c, d). Furthermore, tracer %-MeHg in vegetated soils fell somewhat over the course of the experiment, while %-MeHg in unvegetated soils rose. At the maturity stage, vegetated soils had significantly lower (approximately 35%) tracer %-MeHg than unvegetated soils. Similar, although not significant disparities developed between the ambient %-MeHg levels of vegetated and unvegetated soils at the flowering and maturity stages (Fig. 1c, d, Supplementary Tables 1, 2). These differences between MeHg accumulation in vegetated and unvegetated soils were related to the presence of rice plants rather than loss due to uptake of MeHg to rice tissues, which accounted for only 2–5% of the MeHg mass in the soil of each pot (see Supplementary Text 2 and Supplementary Table 3 for calculations). Given this, our findings can only be mechanistically attributed to a rhizosphere-related enhancement of MeHg demethylation or a slowing of methylation rates that allows background demethylation to degrade the accumulated MeHg. Methylercury demethylation ability is widely distributed throughout the prokaryotes (Lin et al., 2012), and occurs in a wide range of biogeochemical conditions, almost certainly including these experimental pots.

We speculate that plants decrease Hg methylation by reducing the bioavailable fraction of IHg, perhaps through sorption to or co-precipitation with the iron plaque formed on the outside of roots (Meng et al., 2010; Wang et al., 2014) or on root biomass itself (Ullrich et al., 2001). This explanation is supported by the greater effect of vegetation on tracer %-MeHg compared to ambient %-MeHg. As described above, a larger proportion of the ambient IHg was evidently already bound in more recalcitrant, less bioavailable forms, and thus additional decreases in ambient IHg bioavailability would be less apparent in vegetated soils. Alternatively, rice plants may have increased demethylation at later growth stages, perhaps through stimulation of mercury demethylating microorganisms from labile carbon in root exudates.

It should be emphasized that there are many important differences between laboratory microcosms and the field environment, particularly the lack of subsurface porewater transport and its associated solutes, a process that is an important modulator of MeHg dynamics in rice fields (Bachand et al., 2014; Windham-Myers et al., 2014, 2013). These authors observed a rhizoconcentration effect as evapotranspiration drove the movement of porewater MeHg from the bulk to the rhizosphere soil. This process could not occur in our experimental pots, which were entirely colonized with plant roots at the flowering and grain production stages. We hypothesize that the apparent inhibitory effect of plants on soil MeHg accumulation observed in this laboratory experiment, may also occur in the field, but be masked by the rhizoconcentration effect. This suggests that in circumstances where the movement of porewater is very low, plants may decrease, rather than increase MeHg accumulation.

3.3. Uptake, translocation, and bioaccumulation of MeHg and IHg in plant tissues

3.3.1. Methylmercury

The prevailing hypothesis regarding the origin of MeHg in rice grain is that MeHg produced in soil is taken up by roots and translocated to aboveground biomass. This paradigm is supported by correlative evidence linking MeHg concentrations in paddy soils to MeHg concentrations in rice tissues and grain (Meng et al., 2011, 2010; Rothenberg et al., 2011a). The main purpose of the study presented here was to experimentally determine the likelihood of soil as the sole MeHg source to grain using an enriched stable Hg tracer. Alternative sources that would not be detectable in previous work could possibly include adsorption of aqueous or periphyton MeHg through submerged stems or *in planta* methylation of inorganic Hg derived from either soil or the atmosphere (Cosio et al., 2014). In our study, with a unique signal for MeHg derived from soil methylation, a dilution of that signal in plant parts would signify that other sources are important to the accumulation of MeHg in rice.

Our experiment strongly supports the hypothesis that MeHg in rice is solely derived from MeHg produced in the soil where the plant is cultivated. Ambient:tracer ratios for MeHg in aboveground biomass (0.23 ± 0.07–0.34 ± 0.05), including rice grain (0.34 ± 0.05) were close to, but 27% less than ratios observed in soil (0.36 ± 0.04–0.44 ± 0.0 across the three growth stages; Fig. 2a,b, Supplementary Table 2). These differences were statistically significant (Supplementary Table 1). These smaller ambient:tracer MeHg ratios in aboveground biomass could theoretically be related to a bias toward greater demethylation of ambient MeHg *in planta*, *in planta* methylation of tracer IHg, or to preferential translocation of tracer MeHg from soil into aboveground biomass. Previous research has observed that ambient and tracer MeHg are demethylated at equivalent rates (Hintelmann et al., 2000), and that in-planta methylation of IHg in rice is unlikely (Xu et al., 2016). Given these findings, and our consistent observations of greater tracer MeHg compared to ambient MeHg concentrations in plant materials (Fig. 3), our work strongly supports the preferential translocation of newly added tracer MeHg. Moreover, the soil-root bioaccumulation factors, calculated as concentration in roots over concentration in soil, were approximately 28% greater for tracer MeHg than for ambient MeHg (Fig. 2c,d), which matched nearly exactly to the proportionate differences observed in soil versus plant part ambient:tracer MeHg concentration ratios. Similar patterns of increased MeHg bioaccumulation that had been formed from newly added IHg have been previously observed (Mao et al., 2013). MeHg in soils is bound in several pools of varying bioavailability to rice plants (Zhu et al., 2015), and newly formed MeHg may indeed be more available for phyto-uptake. Overall, as others have hypothesized, it is likely that soil MeHg formed in the soil partitions to pore water and enters the roots (Schwesig and Krebs, 2003), travels to the straw via the phloem/symplastic network (Rothenberg

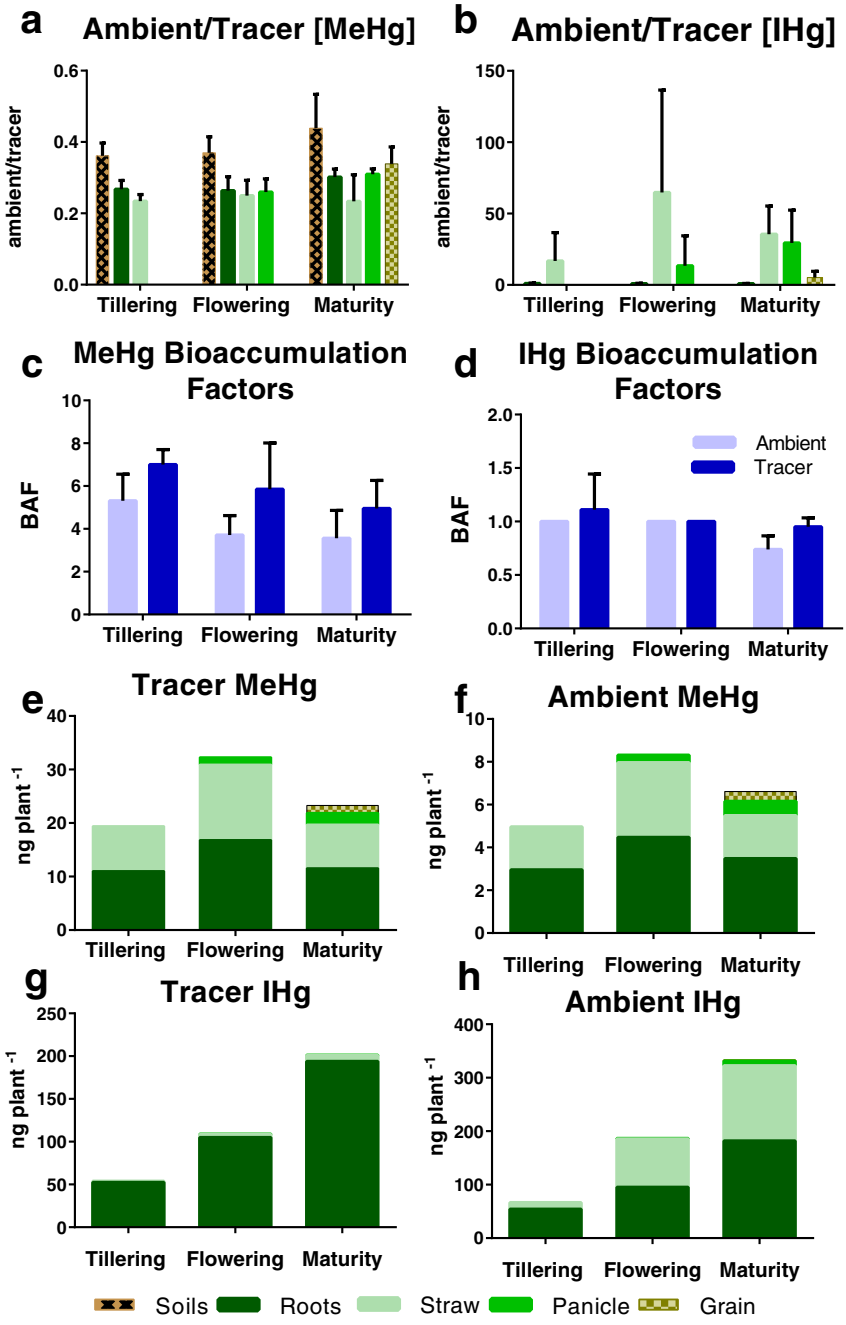


Fig. 2. Accumulation and distribution of ambient and tracer MeHg and IHg in plant tissues at three growth stages. Top: Ambient:tracer ratios (unitless) for MeHg (a) and IHg (b). Middle: Root-soil bioaccumulation factors (unitless) for MeHg (c) and IHg (d). Bottom: Masses of tracer and ambient MeHg (e, f) and IHg (g, h) per plant (ng plant^{-1}). All bars display mean ± standard deviation.

et al., 2015), and then accumulates in the developing grain, possibly as a MeHg-cysteine complex (Meng et al., 2014).

Temporal patterns of MeHg distribution in plant tissues suggest that the greatest uptake of MeHg occurs before flowering, and that growth dilution and *in planta* demethylation may reduce MeHg burdens later in the plant's life. Ambient and tracer MeHg concentrations in roots and straw were highest during tillering, and decreased in subsequent growth stages (Fig. 3, Supplementary Table 2), suggesting that a large initial uptake was diluted by ongoing growth. However, when MeHg concentrations were multiplied by plant biomass to yield the total mass of MeHg in each plant compartment, a different pattern emerged. Ambient and tracer MeHg masses were highest at the flowering stage, increasing significantly (Supplementary Table 1) by approximately 66%, from tillering to flowering (Fig. 2e,f, Supplementary Table 3). These results, which are consistent with previous observations in rice (Meng et al., 2011), indicate that the majority of MeHg uptake occurred prior to the initiation of reproduction (flowering) when vegetative growth was proceeding most quickly. This pattern suggests that investigations into the mechanism of MeHg uptake should focus on nutrient and mineral uptake pathways that are also upregulated at early growth stages, as these may also be involved in MeHg transport. After flowering, we observed that ambient and tracer MeHg masses in roots, straw, and panicles decreased, representing a loss of about 25–30% of the plant's total MeHg burden (Fig. 2e, f, Supplementary Table 2), even after translocation to grain had been accounted for. The only plausible explanation for this net loss is *in planta* demethylation of MeHg during grain development, possibly with further Hg(II) reduction to Hg(0) (Battke et al., 2005) and loss from the plant in the transpirational stream (Lindberg

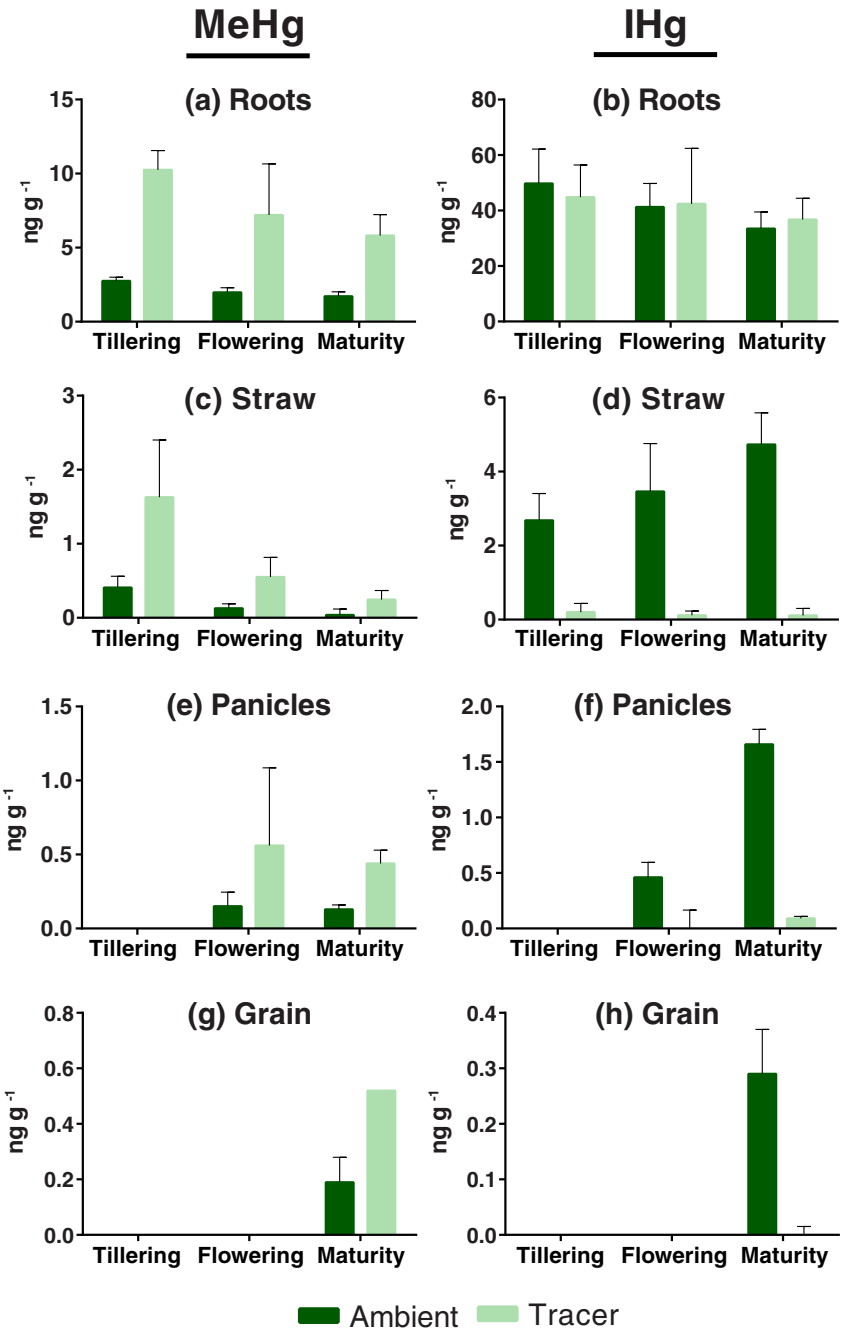


Fig. 3. Ambient and tracer MeHg (left) and IHg (right) concentrations in different plant compartments over three growth stages. All bars display mean ± standard deviation.

et al., 2005). *In planta* demethylation has been observed in living rice plants (Li et al., 2016; Xu et al., 2016), and may be carried out by endophytic demethylating microorganisms or via abiotic, photolytic demethylation (Lin et al., 2012). We speculate that photolytic demethylation, which is effected by UV light, is responsible for *in planta* demethylation, based on the greater losses of MeHg from illuminated straw (approximately 45% lost between flowering and maturity) compared to belowground roots (approximately 22% lost; Supplementary Table 3).

Our stable isotope tracer approach was successfully applied to the question of Hg source; however, the final MeHg concentration in rice grain was lower than expected. MeHg concentrations in grain (ambient: 0.21 ± 0.07 ng g^{-1}, tracer: 0.63 ± 0.21 ng g^{-1}) were below the grain concentrations (1.2–9.4 ng g^{-1}) found by the limited research conducted on rice growing in soils burdened with similar MeHg concentrations (0.5–2 ng g^{-1}) (Li et al., 2013; Meng et al., 2010; Zhou et al., 2015). When MeHg concentrations in rice grain were normalized to THg concentrations, we found that ambient % MeHg values concurred well with other studies (Rothenberg et al., 2014), supporting the ability of our experiment to assess MeHg and IHg sources despite unusually low MeHg concentrations. The cultivar used in this experiment, Nipponbare (also known as Ribenquing in China) accumulates lower concentrations of MeHg than many other varieties (Li et al., 2013). Rothenberg et al. (2012) have further hypothesized that growing conditions, particularly decreased micronutrient availability, may limit MeHg uptake to rice grains. Although plants were actively photosynthesizing and adding biomass throughout the experiment (Supplementary Table 4), growth conditions experienced by our experimental plants were less than optimal, as evidenced by the relatively low grain yields (7% of straw biomass), as well as the final plant biomass (41.57 ± 8.73 g). These findings

support the hypothesis of Rothenberg et al. (2012) that translocation of MeHg to rice grains is partially mediated by growing conditions and/or plant productivity, rather than a simple function of soil MeHg content.

3.3.2. Inorganic mercury

Inorganic Hg in straw and panicles originates partly from the soil and partly from the atmosphere. Tracer IHg was present in the straw and panicles at low but detectable levels (overall mean of 0.27 and 0.11 ng g^{-1}, respectively; Fig. 3c, d and Supplementary Table 2) indicating that translocation of IHg from soil to aboveground tissue does occur. However, foliar uptake was a much more significant source of IHg to rice photosynthetic tissues, contributing 43–100% of the ambient IHg in rice straw (calculated after Mao et al., 2013). Marked and significant disparities were observed between ambient:tracer IHg ratios in straw (overall mean of 39.65) compared to ratios in vegetated soil (1.2) and roots (0.995) (Fig. 2b, Supplementary Tables 1, 2). Foliar uptake of IHg has been previously observed in sawgrass (Mao et al., 2013), and hypothesized in rice (Meng et al., 2010), but this is the first time foliar uptake of IHg has been definitively confirmed in rice. Foliar uptake of Hg from the greenhouse air resulted in an additional burden of approximately 130 ng Hg per plant (Fig. 2h) over the duration of our experiment. High local atmospheric mercury levels can result from industries such as artisanal gold mining or coal combustion, as well as from emissions from highly contaminated soils (Cordy et al., 2011; Eckley et al., 2011; Feng et al., 2002), and may be a significant source of IHg to rice straw in areas where mercury-emitting industries and rice cultivation are in proximity to one another (Meng et al., 2011; Limbong et al., 2003).

Rice roots absorb, but do not bioaccumulate IHg from the soil. The presence of tracer IHg in root biomass confirms the soil origin of this Hg fraction (Fig. 2) (Meng et al., 2012, 2011), but both ambient and tracer IHg concentrations in roots appeared to be in a simple equilibrium with soil concentrations (Supplementary Table 2), with IHg masses increasing over time solely as a result of increasing biomass (Fig. 2h). Supporting this interpretation, bioaccumulation factors for IHg between soil and roots were consistently close to one (Fig. 2d). This contrasts markedly with the observed bioaccumulation of MeHg in roots, supporting the interpretation of other authors that MeHg and IHg are transported from the soil to plant tissue via separate pathways (Rothenberg et al., 2014).

Grain IHg appeared to almost entirely originate from the atmosphere. Ambient IHg was detected in all rice samples (0.30 ± 0.08 ng g^{-1}), but tracer IHg was found at concentrations above detections limits in only two of ten replicates (mean 0.11 ng g^{-1}). Ambient:tracer IHg values for grain (5.25 ± 4.36) were not similar to levels in vegetated soils at the maturity stage (0.94 ± 0.09), but this comparison was necessarily made using the only nonzero values. More convincing is the almost total absence of tracer IHg in rice grain, supporting the interpretation of Meng et al. (2014) that IHg is not translocated from roots, but rather incorporated directly from the atmosphere to rice grains, probably sorbing to protein thiol groups, a moiety with a strong affinity for both MeHg and IHg (Harris et al., 2003). However, it should be noted that translocation of IHg from the soil has been observed in areas with much higher soil IHg concentrations (Feng et al., 2016; Yin et al., 2013), suggesting the sources of IHg to rice grain may vary according to different levels of environmental contamination.

3.4. Conclusions and management implications

This project entailed application of enriched stable isotopes to trace the transformation and movement of IHg and MeHg in a rice-soil system, allowing us to distinguish between soil and atmospheric mercury contamination, and to make suggestions for the practical management of MeHg and IHg accumulation in rice. In plant tissues, comparisons of the ratios of newly added tracer to background ambient MeHg strongly indicate that the MeHg in rice tissues, including grain, originates in the soil. Additional studies using multiple enriched stable isotopes in other plant and environmental compartments would help to conclusively exclude the possibility of *in planta* methylation of inorganic tracer Hg, or the uptake of MeHg from the water via submerged stems (Cosio et al., 2014), but our findings indicate that these are most likely very minor if they occur at all. While our findings should be corroborated by similar experiments in working paddy rice fields, our observations do suggest several avenues for the practical management of MeHg and IHg accumulation in rice. In examining patterns of uptake of MeHg over the course of a plant's life, the majority of MeHg is absorbed before flowering, suggesting that mitigation measures such as paddy water drawdown (Peng et al., 2012), should be timed for this interval. Losses of the total mass of MeHg from individual plants between the flowering and maturity stages further indicate probable *in planta* demethylation in rice, the timing of which should be further assessed as a potential strategy for phytoremediation of soil or limitation of MeHg uptake to grain.

In contrast to MeHg, IHg in rice tissues originates from at least two sources, presenting different management challenges. Root burdens of IHg derive solely from soil uptake, while the IHg in rice straw is absorbed partially from the atmosphere, and grain IHg burdens are absorbed almost completely from the ambient air. The common practice of reincorporating rice straw to fields may inadvertently increase Hg contamination of agricultural lands by transferring atmospheric Hg to the soil. This is a particular concern in areas with high atmospheric Hg concentrations and thus large potential for foliar uptake of IHg. This new IHg input may stimulate mercury methylation, and thus explain observed increases in the production and uptake of MeHg in fields amended with rice straw (Liu et al., 2016). In grain, the direct absorption of IHg from the atmosphere rather than uptake from soil means that IHg concentrations in rice are likely to be little affected by any regulatory measure short of regional reductions in atmospheric Hg. This is an important food safety consideration since, unlike other foods (Munthe et al., 2007), IHg makes up 60–70% of the total Hg in rice grains according to a recent review by Rothenberg et al. (2014).

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Appendix A. Supplementary data

Further information on the results of statistical tests; summary statistics of selected variables; calculations for MeHg mass-balance in soil and rice plants in vegetated pots; estimations of potential Hg flux from experimental pots; and data on plant biomass and photosynthetic rate. Supplementary data associated with this article can be found in the online version, at http://dx.doi.org/10.1016/j.scitotenv.2016.08.068.

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